

# Elizabeth Vargas

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| PROFESSIONAL EXPERIENCE | <b>Leap AI</b><br><i>Senior Machine Learning Engineer</i> | Jan. 2023 - Present<br>Edinburgh, United Kingdom |
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- Computer vision and machine learning in a robotic system for product packing.
- Develop and deploy novel computer vision formulations, models, and algorithms for object segmentation, classification, and tracking (PyTorch/OpenCV).
- Design experiments for methodical data collection, analysis and improvement of vision pipelines, ensuring reliable performance in varied operational conditions.
- Resolve issues related to AI/ML model performance and infrastructure (Docker).
- Work closely in cross-functional teams, ensuring the vision system's seamless integration and performance in the broader robot system (ROS/Git).

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|  | <b>Ouster Automotive</b><br><i>Software Engineer</i> | Dec. 2020 - Oct. 2022<br>Edinburgh, United Kingdom |
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- Ouster acquired *Sense Photonics* in 2021 and established *Ouster Automotive*.
- Performed statistical modelling and data analysis to support the design of 3D Flash LiDAR to be deployed within self-driving vehicles.
- Developed a simulation software that matches real world measurements employed to guide LiDAR design specifications (Python, Scipy, C++).
- Defined and perform experiments to test the performance boundaries of signal processing algorithms used to process data from the LiDAR system.

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|  | <b>ORCA Hub</b><br><i>Research Associate</i> | Jan. 2019 - Nov. 2020<br>Edinburgh, United Kingdom |
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- Developed a real-time computer vision system, deployed on a remotely operated underwater vehicle for the surveying and inspection of offshore assets (ROS, C++).
- Improved sensor calibration process to ensure precise alignment and synchronisation of multi-sensor data.
- Implemented Visual Simultaneous Localisation And Mapping (SLAM) solution for limited visibility environments, fusing data from acoustic and optical sensors.
- Employed stereo cameras for 3D reconstruction of submerged structures, enabling the offshore industry to inspect and certify their integrity.
- Designed, set-up, and executed practical experiments and prototype systems relating to the demonstration of solution capabilities to industry representatives.

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|  | <b>Toshiba Medical Visualization Systems</b><br><i>Research Intern</i> | Jun. 2015 - Sep. 2015<br>Edinburgh, United Kingdom |
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- Characterised Alzheimer using Magnetic Resonance Imaging (MRI), performing texture analysis in brain tissue enabling the early diagnose of the disease.
- Combined brain gyrus segmentation with regional texture metrics (Pandas).
- Applied supervised machine learning techniques to image texture, including feature selection, classification and regression (Python, Scikit-Learn).

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| EDUCATION             | <b>Ph.D. Signal Processing</b> Oct. 2015 - Sep. 2019<br><i>Heriot-Watt University, United Kingdom</i> <ul style="list-style-type: none"> <li>Advanced the state-of-the-art in acoustic source localisation in constrained environments through three major contributions (detailed below).</li> <li>Reduced computation six fold while maintaining localisation accuracy at state-of-the-art levels (Python, NumPy, SciPy).</li> <li>Implemented a signal sampling algorithm to achieve accurate localisation for a signal transmitted at a compression ratio of 40 : 1 (MATLAB).</li> <li>Applied deep learning techniques to achieve a 20% improvement in localisation accuracy using data augmentation from a GAN (Python, Keras, TensorFlow).</li> </ul> |
|                       | <b>M.Sc. Computer Vision and Robotics with Distinction</b> Sep. 2013 - Jun. 2015<br><i>University of Burgundy, France</i> GPA: 15.3/20 <ul style="list-style-type: none"> <li>Joint Erasmus Mundus Master Program with <i>University of Burgundy</i> (France), <i>University of Girona</i> (Spain) and <i>Heriot-Watt University</i> (United Kingdom).</li> <li>Basis of signal and image processing, medical image analysis (MATLAB).</li> <li>Segmentation, multi-view geometry, object recognition and tracking (OpenCV).</li> <li>Robot autonomy and intelligence, including SLAM and motion planning (ROS).</li> </ul>                                                                                                                                  |
|                       | <b>B.Sc. Computer Science</b> Aug. 2006 - Aug. 2012<br><i>Universidad del Valle, Colombia</i> GPA: 4.67/5.0 <ul style="list-style-type: none"> <li>Courses in algorithms, data structures, compilers and software engineering.</li> <li>Projects including image processing (C/C++), search algorithms, optimisation, evolutionary algorithms, software development (Java) and databases (MySQL).</li> </ul>                                                                                                                                                                                                                                                                                                                                                 |
| SELECTED PUBLICATIONS | <b>E. Vargas</b> , R. Scona, J. Scharff Wilners, T. Luczynski, Y. Cao, S. Wang, Y. Petillot, <i>Robust Underwater Visual SLAM Fusing Acoustic Sensing</i> , in International Conference on Robotics and Automation ( <b>ICRA</b> ), Xina, China, June 2021.<br><br><b>E. Vargas</b> , J. R. Hopgood, K. Brown, K. Subr, <i>On Improved Training of CNN for Acoustic Source Localisation</i> , in Transactions on Audio, Speech, and Language Processing ( <b>TASLP</b> ), 2021.                                                                                                                                                                                                                                                                              |
| DISTINCTIONS          | <b>Erasmus Mundus Scholarship</b> , <i>European Commission</i> Sep. 2013<br>Granted to 4 European students for academic and professional achievement to study a Master in Computer Vision and Robotics (ViBot) during the academic year 2013-2015.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| TRAINING              | <b>International Summer School on Deep Learning</b> Jul. 2018<br>Research training event aiming at updating participants about the most recent advances in the critical and fast developing area of deep learning.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                       | <b>International Computer Vision Summer School (ICVSS)</b> Jul. 2016<br>Provided an objective, clear, and in-depth summary of the state-of-the-art research in the areas of Computer Vision, Machine Learning and Artificial Intelligence.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| TECHNICAL SKILLS      | <b>Operating Systems:</b> Windows, Linux (Ubuntu)<br><b>Programming Languages:</b> Python, MATLAB, C/C++<br><b>Robotics:</b> Robotics Operating System (ROS), ROS2, Gazebo, Rviz<br><b>Computer Vision:</b> OpenCV, Scikit-image<br><b>Machine Learning:</b> Numpy, Scipy, Scikit-learn, Pandas, PyTorch<br><b>DevOps:</b> Git, Docker<br><b>Agile Tools:</b> Jira, Confluence                                                                                                                                                                                                                                                                                                                                                                               |